PV102 Day 4 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 20 | Morning Quiz - Answer Key
String design follows limits, temperature, and shade reality. | Thursday, August 6, 2026

Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. In series wiring, matching module voltage primarily:

- A. Adds
- B. Disappears
- C. Becomes AC
- D. Turns into resistance

Answer: A

Rationale/look-for: Series wiring adds voltage while current is limited by the string.

2. In parallel wiring, matching branch current primarily:

- A. Adds
- B. Becomes zero
- C. Changes to frequency
- D. Removes voltage

Answer: A

Rationale/look-for: Parallel wiring adds current while voltage remains at branch voltage.

3. The charge controller input limit is important because:

- A. Exceeding limits can damage equipment or create unsafe conditions
- B. It chooses wire color
- C. It makes sunlight brighter
- D. It stores energy

Answer: A

Rationale/look-for: Equipment limits are hard design boundaries.

4. Shade on one module can affect:

- A. Only the paper worksheet
- B. Production and string behavior
- C. Lunch time only
- D. Nothing if the roof is new

Answer: B

Rationale/look-for: Shade reduces production and can create mismatch effects in connected circuits.

5. Cold temperature generally makes PV voltage:

- A. Higher
- B. Always zero
- C. AC only
- D. Unrelated to design

Answer: A

Rationale/look-for: Cold conditions can raise PV voltage and matter for maximum input limits.

6. Why should string design be checked before connecting equipment?

_____

_____

Answer: Safety and limits

Rationale/look-for: Students should mention voltage/current limits and equipment protection.

7. A controller allows 100 V max input. Should a design near 99 V be accepted without temperature check?

_____

_____

Answer: No

Rationale/look-for: Cold-weather voltage margin must be checked.

8. Name one reason parallel strings may require added protection.

_____

_____

Answer: Backfeed/current

Rationale/look-for: Parallel sources can increase fault current and may require OCPD depending on system design.

9. What evidence from Day 2 matters most for shade design?

_____

_____

Answer: Shade observations

Rationale/look-for: Shade timing, object location, and candidate array area.

10. Write the Day 4 rule in your own words.

_____

_____

Answer: Limits and shade

Rationale/look-for: Look for limits, temperature, shade, or electrical layout.